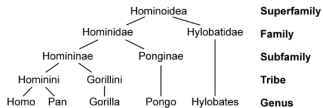




This family tree shows the evolutionary relationships between humans (Homo) and some of their living relatives.



Identify the classification of Gorilla in the family tree.

Gorillini

Hominini

Hominidae

Homininae

Pan

Hominoidea

Gorilla belongs to the family of , the subfamily of , and the tribe of .





The photos show a hot water bottle and a daisy wheel pattern on it that indicates the week, month and year of manufacture. The daisy wheel has 12 segments, each representing one month of the year 2018.

January is the top segment with dots and other months follow in a clockwise direction.

Each dot in the segments represents one week.

The last segment containing dots indicates the week and the month that the bottle was made.



Move the text to correctly complete the sentence.

second

third

eleventh

February

March

November

The bottle was made in the week of , 2018.



A sparkler is a metal wire coated with a grey chemical. The video shows that the grey chemical sparkles as it burns.



Gavin breaks off the grey chemical from two sparklers and puts each in a separate pile, as shown in the image.



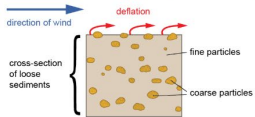
He then times how long each pile takes to burn.

What factor has Gavin chosen to change in his investigation?

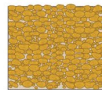
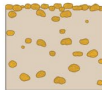
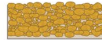
- ☐ the amount of grey chemical to burn
- ☐ the shape of the pile of grey chemical
- ☐ the length of time each pile takes to burn
- ☐ the size of the pieces of grey chemical



Deflation is when wind removes fine particles from a surface of loose sediments but leaves behind coarse particles. This is the most important form of erosion in flat, dry deserts consisting of loose sediments.



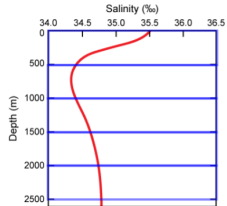
What is the most likely result of deflation of this cross-section of loose sediments over a long period of time?





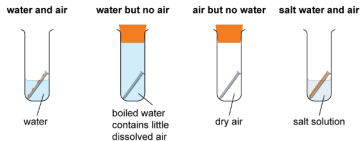
Salinity is a measure of the amount of salt in water or soil. It is usually given in parts per thousand (‰). A halocline is a layer of seawater  in which salinity changes rapidly with depth.

Select the region of the graph that best represents a halocline.





Sam carried out an investigation into rusting. The diagram shows what happened to iron nails placed in four test tubes exposed to different conditions and left for several days.

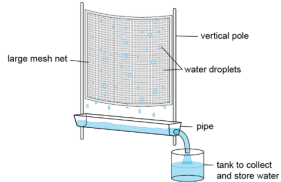


Which of the following is an observation based on Sam's investigation?

- ☐ The rusting of iron nails requires a combination of oxygen and water.
- ☐ The rusting of iron nails is speeded up by the addition of salt.
- ☐ Orange flakes form on iron nails in the open tube containing water.
- ☐ Orange flakes will form on iron nails in a tube filled with humid air.



Fresh water can be collected in a variety of ways. The diagram shows a way of water collection using a large mesh net suspended between two vertical poles. Droplets of water collected on the net join together and flow down into a pipe below the mesh net, and then into a collecting tank.



These nets are most likely designed to collect water from



dew.



fog.



rain.



snow.



The human body radiates energy as infrared radiation. A visible light photograph and an infrared photograph are shown below.



Move the text boxes to complete the following sentence.

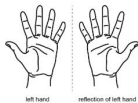
infrared radiation

visible light

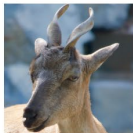
The plastic sheet is transparent to

The man's glasses are opaque to

When the left hand is held up to the mirror, its reflection in the mirror looks like the right hand. This interchange of images is called lateral inversion.



When held up to face a mirror, which one of these animals would likely **not** display a lateral inversion?





Rebecca has the following equipment.

plastic bottle



jug with water



wooden board



wooden blocks



tape measure



notepad and pen



Which question **cannot** be answered using only the listed equipment?

- ☐ Does the type of bottle affect how far it will roll?
- ☐ Does the amount of water in a bottle affect how far it will roll?
- ☐ Does the steepness of a ramp affect how far a bottle will roll?
- ☐ Does the wooden board break when supporting a full bottle of water?

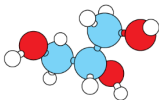


The molecular formula of a compound gives the number of atoms of each element in a single molecule.

The molecular formula for water is H_2O . This indicates one molecule of water has two hydrogen atoms and one oxygen atom. The diagram shows the structure of water.



The molecule below is made of three types of atoms: carbon, oxygen and hydrogen.



Which of the following is the correct molecular formula for this compound?



- ☐ $\text{C}_3\text{H}_8\text{O}_3$
- ☐ $\text{C}_3\text{H}_3\text{O}_8$
- ☐ $\text{C}_3\text{H}_5\text{O}_3$
- ☐ $\text{C}_3\text{H}_8\text{O}_2$



Jane investigated the effect of different types of red light ('red' light and 'far-red' light) on the germination of lettuce seeds.

She placed lettuce seeds on moist paper in five dishes. She then placed the dishes in a dark cupboard and kept them there.

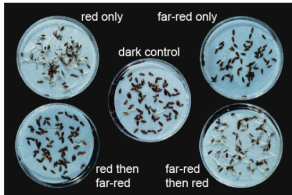
She kept one dish in the dark at all times.

Each day she exposed the other four dishes to different patterns of flashes of red light:

- red light only;
- far-red light only;
- red light followed by far-red light; or
- far-red light followed by red light.

Immediately after the light exposure, the four dishes were returned to the dark cupboard.

The photograph shows her dishes after a few days.



The table shows Jane's observations, but some information is missing.

Drag the missing information for Dish 4 into the table.

red then far-red

far-red then red

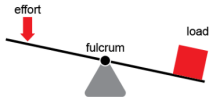
a few grew

a lot grew

	Light used	Seed growth
Dish 1	dark control	a few grew
Dish 2	red only	a lot grew
Dish 3	far-red only	a few grew
Dish 4		
Dish 5		a lot grew



The diagram shows a simple lever.

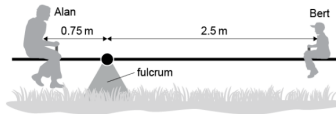


A lever is balanced when the following condition is met:

effort × distance between effort and fulcrum = load × distance between load and fulcrum

A see-saw is an example of a lever.

The see-saw in the diagram is balanced. Alan has a mass of 100 kg. He is sitting 0.75 metres from the fulcrum. Bert is sitting 2.5 metres from the fulcrum.



What is Bert's mass?

☐

56.3 kg

☐

53.3 kg

☐

40.0 kg

☐

30.0 kg





Mick had six oranges, each weighing about 100 g. He placed one orange in a jug of water, as shown.



water level without the orange

He also had a 9-litre bucket containing 5 litres (L) of water. He placed all six oranges into the bucket and then poured another 5 L of water into the bucket.

Approximately how much water flowed out of the bucket?



0.6 L



1.6 L



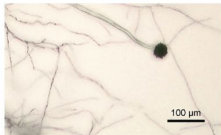
2.8 L



3.4 L



Anne found a micrograph of the fungus *Aspergillus niger* in a book.



The length of the scale bar in the book was measured to be 2 centimetres (cm). One millimetre (mm) equals 1000 micrometres (μm).

What is the magnification of this micrograph in Anne's book?

☐

500 times

☐

200 times

☐

50 times

☐

20 times



A bull's eye level contains fluid and a gas bubble. When it is on a level surface, the bubble is in the centre of the marked circle.



The diagram shows a bull's eye level on a wooden tray viewed from above.

Select **all** the corners that need to be lifted slightly to make the tray level.





Don grows garlic on his farm. He has planted three different varieties of garlic, in rows, in two test beds, as shown in the table.

Test bed 1 - Soil				Test bed 2 - Compost		
row 1	row 2	row 3	row 4	row 1	row 2	row 3
purple stripe	purple stripe	porcelain	porcelain	purple stripe	silverskin	silverskin

As Don harvests each row, he weighs its crop to calculate the yield.

Which of the following **two** questions can he answer with his data?

☐

Which variety has a better yield, silverskin or porcelain?

☐

Does porcelain have a better yield in soil than in compost?

☐

Does purple stripe have a better yield in compost than in soil?

☐

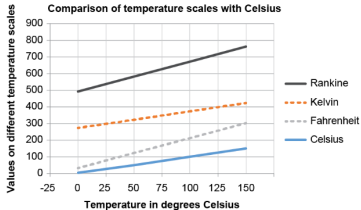
Which variety has a better yield, purple stripe or porcelain?

☐

Does compost help silverskin to have a better yield?

There are a number of different scales that can be used to record temperature.

The graph shows how four different temperature scales are related.



Water freezes and becomes a solid at 0 degrees on the Celsius scale. Water boils, turning to vapour, at 100 degrees on the Celsius scale.

At approximately what temperature does water boil on the Rankine scale?

- ☐ 100 degrees
- ☐ 500 degrees
- ☐ 670 degrees
- ☐ 750 degrees



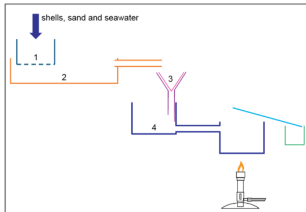
There are a number of processes by which components of a mixture can be separated. Some of them are described below.

Filtration is the process of using a filter to separate a solid from a liquid.

Sedimentation is the settling out of a suspended material.

Sieving is the mechanical process of separating out particles based on size.

Julia wanted to collect salt crystals from a mixture of large shells, sand and seawater. She used the processes in the diagram to achieve her aim.



Where was the sedimentation process used?



1



2



3



4



Tina found a recipe for making pie dough.

pie dough recipe

240 g flour

145 g shortening

30 g butter

60 g water

2 g salt



Butter contains about 15% water and 85% fat, by mass. Shortening contains 100% fat.

Tina wanted to make pie dough without butter.

How much shortening should she use to keep her recipe consistent with the recipe she found?

g



Chemiluminescence is emission of light as a result of a chemical reaction. For example, the reaction of luminol and hydrogen peroxide, catalysed by metal ions, forms an excited intermediate that then gives off energy as blue light as the intermediate turns into a product of a lower energy level. The reaction can be represented by the chemical equation:



Forensic scientists use this reaction to detect blood at crime scenes. A mixture of luminol and hydrogen peroxide solution is sprayed onto a surface. Any area with traces of blood will glow with blue light.

Which of the following statements is correct about the application of this reaction in forensic science?

- ☐ Blood contains luminol that reacts with the chemical spray.
- ☐ Blood contains metal ions that react with the excited intermediate.
- ☐ Water and oxygen in the blood form H_2O_2 that reacts with luminol.
- ☐ Metal ions in the blood facilitate the reaction of luminol and H_2O_2 .





Cathy and Jeff timed how long it took them to swim each of two laps in a swimming pool.

Their results are summarised in the table.

Swimmer	Time for lap 1 (mm:ss)	Time for lap 2 (mm:ss)
Cathy	00:57	01:05
Jeff	00:54	01:03

Who was the first to complete the two laps and what was the time difference between the swimmers?

- ☐ Cathy with a time difference of three seconds
- ☐ Jeff with a time difference of five seconds
- ☐ Cathy with a time difference of five seconds
- ☐ Jeff with a time difference of two seconds



Peter dropped two identical metal balls from different heights into a block of soft clay.

The diagram shows what a cut section of the block looked like after both balls stopped moving downwards into the clay.



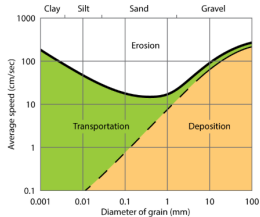
Which of the statements is correct?

- ☐ The two balls had different amounts of kinetic energy before they were dropped.
- ☐ Ball 1 hit the surface of the clay block with less kinetic energy than ball 2 did.
- ☐ Ball 2 hit the surface of the clay block with more potential energy than ball 1 did.
- ☐ The two balls lost all potential energy when they stopped in the clay block.



The speed of the current and the size of sediment particles at the bottom of a water body determine what happens to these particles. Erosion occurs when sediment particles are forced to break away from their position. These particles can then be transported to another place by the moving current. Deposition occurs when the particles carried by the current sink back to the bottom.

This graph summarises the relationship between particle size, current speed and the processes of erosion, transportation and deposition.



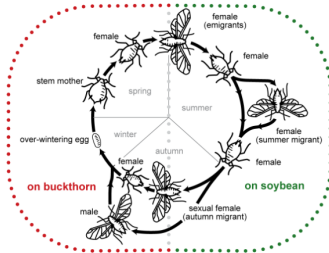
Which of these statements is supported by the information in the graph?

- ☐ A current with a speed as low as 0.01 cm/sec can transport the eroded silt particles.
- ☐ Gravel particles can be transported and deposited in a current with a speed of 100 cm/sec.
- ☐ The larger the sediment particle, the greater current speed is needed to erode it.
- ☐ Clay can be eroded by a current with a speed lower than is needed to erode sand.



The soybean aphid is a common pest of soybeans. It has a complicated life cycle involving two host plants, buckthorn and soybean, and both sexual and asexual reproduction.

The diagram shows the reproduction cycle of a type of soybean aphid. Each insect in the diagram represents one generation.

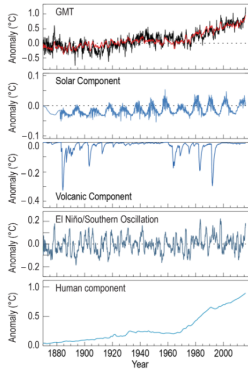


Which **two** statements are supported by the information in the diagram?

- ☐ Wingless females produce winged females only.
- ☐ Sexual reproduction occurs only on buckthorn.
- ☐ Both winged and wingless females produce offspring.
- ☐ Winged aphids move from buckthorn to soybean plants in autumn.



This set of graphs show the estimated contribution of various natural and human factors to change in global mean temperature (GMT) from 1870 to 2015. In atmospheric science, an anomaly is a deviation in quantity from an expected value.

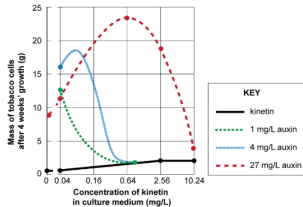


Which of the following statements is supported by the information in the graphs?

- ☐ The fluctuation of GMT was not affected by human factors.
- ☐ Human factors had a greater effect on GMT than did all natural factors.
- ☐ Human factors started affecting GMT from 1970.
- ☐ The GMT fluctuated exactly the same way as El Niño/Southern Oscillation.



This graph shows how plant hormones, auxin and kinetin, affect the growth of tobacco cells in culture. The hormones were added to the culture medium on which the cells were growing.



What is the dependent variable in this investigation?

- ☐ the concentration of kinetin in the medium
- ☐ the concentration of auxin in the medium
- ☐ the time the cells took to reach optimum growth
- ☐ the mass of tobacco cells after 4 weeks

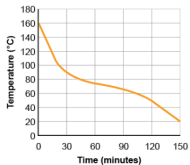
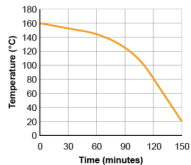
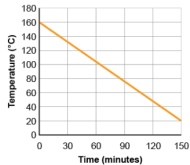
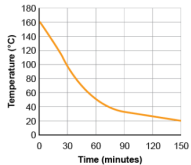




The rate at which a hot object cools down is proportional to the difference between the temperature of the object and the temperature of its surroundings.

The temperature in Andre's kitchen was 20°C . Andre baked a pie in the oven. He took the pie out of the oven and put it on the kitchen table to cool down for 150 minutes. He recorded the temperature of the pie as it cooled using a data logger.

Which graph best represents the temperature of the pie as it cooled?





The 3-D arrangement of atoms within a molecule can be drawn in 2-D using conventional symbols for bonds:

a bond that aligns with the plane of the screen is drawn as a straight line

a bond that points away from you is drawn as a dashed line

a bond that points towards you is drawn as a wedged line.

Three of the following diagrams represent the same molecule. Which molecule is different?

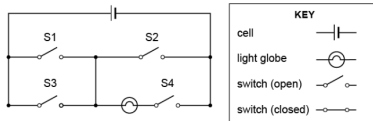




Switches control the flow of electricity around a circuit. A closed switch allows the current through and turns the circuit on. An open switch breaks the path for the electricity and turns the circuit off.



Dave made this circuit.



Initially all the switches were open and the light was off.

Dave closed two switches and the light came on.

Which **two** combinations of switches could Dave have closed to achieve this result?

☐

S1 and S4

☐

S2 and S3

☐

S2 and S4

☐

S3 and S4

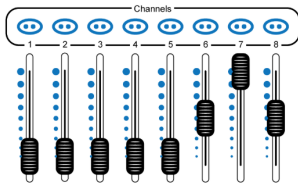




Two 500 W PAR can lights and a 1000 W spotlight are connected to a dimmer rack.



The dimmer rack is connected to lighting channels 6-8 in the order as the lights shown above.



What will be the total power drawn by the three lights?



750 W



1000 W



1250 W



1500 W



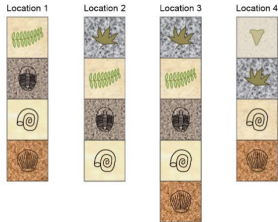
2000 W



Fossils are found in sedimentary rocks. Sedimentary rocks form in layers. The layer closest to the surface is usually the youngest layer of rock, so it contains the youngest fossils.



The diagrams show columns of rock taken from four locations in the same area.



Which layer of rock is the youngest?

☐☐☐☐



A lightning bolt is a discharge of electricity that heats the air surrounding it. This heated air expands rapidly, which we hear as thunder. We see a bolt of lightning before we hear the thunder from it. The sound of thunder takes about 5 seconds to travel a mile, and about 3 seconds to travel a kilometre.



The following students saw a bolt of lightning:

Nick, who was 2 miles away from the lightning bolt

Simon, who counted 8 seconds between seeing the lightning bolt and hearing the thunder

Ros, who was 3 kilometres away from the lightning bolt.

Move the names to correctly complete the sentence.

Simon

Nick

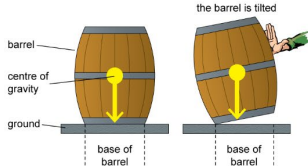
Ros

was closer than to the lightning bolt, but further away than .



The centre of gravity of an object is the point in the object through which the weight of an object appears to act.

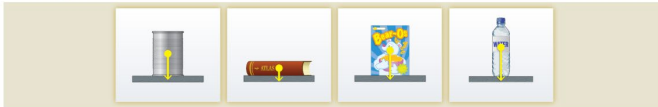
The diagram shows that a vertical line can be drawn from an object's centre of gravity to the ground. When that object is tilted, the relative position of the vertical line to the ground will shift.



An object will topple over if tilting the object causes the vertical line from its centre of gravity to shift to outside the base of the object.

Each of the following objects is slowly tilted towards the left at the same speed, without any slipping.

Move the objects to order them from first to last to topple over.



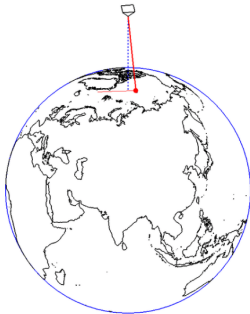
first

last



A pendulum consists of an object, called a bob, hanging by string from a fixed point. When pushed, the bob will swing back and forth.

The video shows a large pendulum placed at the North Pole viewed from space. The path of the swinging bob over the ground is marked by a red line. The movement of the path may appear different for observers from space and on the ground.



Another identical pendulum is positioned at the equator.



Complete the table to describe the path of the swinging bob to an observer on the ground at each location.

clockwise rotation

anticlockwise rotation

no rotation

24 hours

more than 24 hours

12 hours

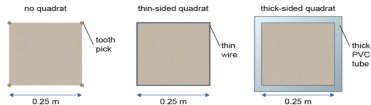
0 hour

	Movement of the path	Time taken to return to its starting position
At North Pole		
At Equator		

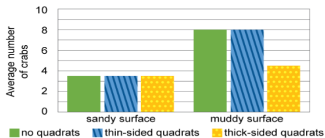


Fiddler crabs emerge from burrows at low tide to feed on the surface. Scientists estimate the number of crabs living in these habitats by counting how many emerge within small sampling areas during a 10-minute period. The sampling areas were delineated by quadrats which are frames of a specific size.

Maddy performed an experiment to determine how two types of quadrats affect counts of these crabs on a sandy surface, as shown.



She then repeated the experiment on a muddy surface. The graph shows Maddy's results.

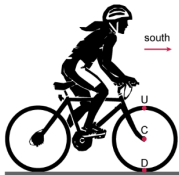


Which **two** statements are supported by the results?

- ☐ Quadrats only give accurate counts on a sandy surface.
- ☐ Thin-sided quadrats give accurate counts on both surfaces.
- ☐ Some crabs avoid thick-sided quadrats on a muddy surface.
- ☐ Both thin-sided and thick-sided quadrats give accurate counts on both surfaces.
- ☐ Some crabs are attracted to thin-sided quadrats on a muddy surface.



Liz was riding her bicycle on a straight road at a constant speed of 20 km/h, heading south.



At a given time, a person on the footpath watched her passing.

Select the instantaneous speed and direction of the three points (U, C and D) on the wheel relative to the footpath observer.

0 km/h

20 km/h north

20 km/h south

40 km/h north

40 km/h south

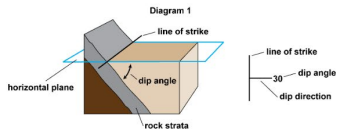
Point U

Point C

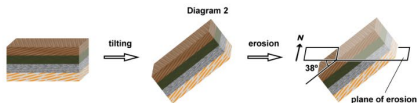
Point D



In geological maps, 'strike' is the direction of the line formed by the intersection of a rock surface with a horizontal plane; 'dip' is the acute angle that a tilted rock surface makes with a horizontal plane. A slope dipping downwards in a certain direction can be represented by a 'strike and dip' symbol, as shown in Diagram 1.



Over millions of years of geological change, the rock strata in an area were tilted and eroded to form a horizontal flat surface, as shown in Diagram 2.



Geologists often use a top view map of the erosion plane and strike and dip symbols to describe the tilted rock strata of an area. Which of the following are the correct top view map and the strike and dip symbol for the eroded rock strata in Diagram 2?

N
↑



N
↑



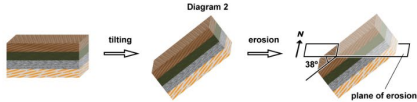
N
↑



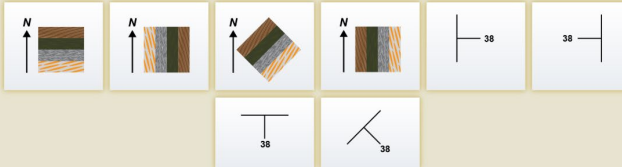
N
↑



Over millions of years of geological change, the rock strata in an area were tilted and eroded to form a horizontal flat surface, as shown in Diagram 2.



Geologists often use a top view map of the erosion plane and strike and dip symbols to describe the tilted rock strata of an area. Which of the following are the correct top view map and the strike and dip symbol for the eroded rock strata in Diagram 2?



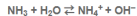
Map of top view

Strike and dip symbol



Ammonia (NH_3) is highly soluble in water. A high concentration of ammonia causes direct toxic effects on aquatic life.

When ammonia dissolves in water, some of it is converted to the nontoxic ammonium ion (NH_4^+) according to the equation:



The percentage of ammonia in a solution (A%) changes with pH and temperature, as shown in the table below.

Temperature (°C)	pH 6.5	pH 7	pH 7.5
15	0.0865	0.273	0.858
20	0.125	0.396	1.24
25	0.180	0.566	1.77
30	0.255	0.801	1.49

The Total ammonia-N (TAN) represents the total concentration of nitrogen (N) in the form of ammonia and ammonium ion in a solution. It is used to measure the toxicity of ammonia in water bodies, and can be calculated using the formula:

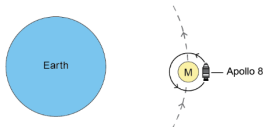
$$\text{TAN} = \text{NH}_3 \text{ concentration in } \mu\text{g/L} \times \frac{14}{17} \times \frac{100}{\text{A}\%}$$

For a solution with an NH_3 concentration of 68.7 $\mu\text{g/L}$ at pH 7 and temperature 25°C, what is the Total ammonia-N, in **mg/L**, rounded to one decimal place? 1 mg = 1000 μg .

mg/L



In December 1968, the crewed spacecraft Apollo 8 was launched to orbit the Moon, as shown.



At one point when the camera on the spacecraft was tilted to the horizon of the Moon, astronaut William Anders witnessed an 'Earthrise' and took this famous photo.



Select the most likely position of the Moon relative to the Sun and Earth when the photo was taken.

